Section A

Cluster 2 Fieldwork

1 A group of four Junior College students wanted to investigate how community-led initiatives in Singapore contribute to climate change mitigation by addressing the "buy-and-throw away" culture. They focused on the Repair Kopitiam initiative, which aims to reduce waste and carbon footprints associated with the production of new goods. Volunteer coaches guide members of the public to fix their own broken items.

The students conducted their investigation at Our Tampines Hub, an integrated community hub where monthly repair sessions are held. Their aim was to determine if such community-led sessions effectively encourage residents to adopt sustainable consumption habits which would help to mitigate climate change.

The students' geographical question was:

"How does awareness of Repair Kopitiam influence the frequency of climate-mitigating repair behaviours among users of Our Tampines Hub?"

The students conducted their data collection on a Sunday between 12:00 pm and 2:00 pm to coincide with a scheduled Repair Kopitiam session at Our Tampines Hub.

They identified the Festive Plaza – the large central sheltered atrium on the ground floor – as the hub's main meeting point for users and conducted their survey there. Using convenience sampling, two of the students approached passers-by within the Festive Plaza to complete their questionnaire. They aimed for a target sample size of 50 users but collected a total of 22 completed responses at the end of the 2-hour period.

The other two students stationed themselves at the Repair Kopitiam coaching stations to take photographs and observe the type of items that were brought for repair. The Repair Kopitiam coaching stations were located at Festive Drive, a separate sheltered walkway and driveway area situated along the perimeter of Our Tampines Hub. This was a 1 to 5-minute walk away from the Festive Plaza, depending on the crowd.

Resource 1 shows photographs of the Festive Plaza and the Repair Kopitiam coaching stations. Resource 2 shows the questionnaire that the students designed to conduct their survey. Resource 3 is a scatter graph of the responses to Question 6 and Question 9 of the questionnaire survey.

(a) Explain why the geographical question is appropriate for this investigation. [4]

Award 1 mark for each explanation of why the GQ is appropriate, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- It is at a **suitable scale** as the question is localized to a single, manageable spatial unit – Our Tampines Hub. [1 mark]
    - There are only 4 students carrying out the investigation and pursuing the study at more sites will not be manageable for the student. [1 additional mark]

- **Specific and Clearly Defined**
    - It is clearly defined as it identified the specific target group which are the users of Our Tampines Hub. [1 mark]

    - It specifies a community-led initiative that is happening within Our Tampines Hub – "Repair Kopitiam" – instead of keeping it open. [1 mark]
        - If the initiative was not specified, it might have been difficult to attribute a reason for any changes in consumption habits of users. [1 additional mark].

- It is **researchable**/ **measurable** because it is possible for the students to collect the required data about awareness of Repair Kopitiam and/or the frequency of climate-mitigating repair behaviours. [1 mark]
    - Both variables in the question can be quantified using standard fieldwork methods such as interviews or surveys. [1 additional mark]

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(b) With reference to Resource 1, explain how the students could minimise the potential risks and impacts of their investigation at Our Tampines Hub. [6]

Award 1 mark for each suggestion on how the students could minimise potential risks or impacts during their investigation, to a maximum of 6 marks.

Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Award a maximum of 4 marks for suggestions of how the students could minimise either risks or impacts only.

Possible responses include:

Minimising risks

- (Resource 1A) Students could keep their personal belongings with them or within site while they are administering the questionnaire survey to avoid any loss in view of the relatively crowded conditions. [1 mark]
- (Resource 1B – Evidence A) Students could be mindful of not tripping on any wires lying on the ground around the coaching stations. [1 mark]
- (Resource 1B – Evidence B/C) Students could keep a safe distance from the Repair Kopitiam coaching stations while they are taking photos of the repair activity to avoid being injured by the electrical equipment or sharp objects. [1 mark]
    - They could also avoid touching any of the exposed wires or circuit boards to avoid electrocution. [1 additional mark]

Minimising impacts

- (Resource 1A) Students could choose a less crowded area of Festive Plaza to administer the questionnaire survey so as not to get in the way of users of Our Tampines Hub and inconveniencing them. [1 mark]
    - This is especially so for users who are handling prams, as seen in the left foreground of Resource 1A, who may need more space for navigation. [1 additional mark]
- (Resource 1A) Students could avoid administering the questionnaire survey near/ at the entrance of commercial establishments so as not to hinder customers from entering and disrupting the business. [1 mark]
- (Resource 1B – Evidence B/C) Students could avoid being in the way of the repair coaches while they are taking the photographs so as not to disrupt the coaching process. [1 mark]
    - This is especially so since it would be necessary for users to have a good view of how the volunteer coaches are carrying out the repair so as to learn how to do the repairs on their own.

(c) With reference to Resource 2, explain how the design of the questionnaire could be improved. [6]

Award 1 mark for each explanation on how the design of the questionnaire could be improved, to a maximum of 6 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- The fields for "Full Name" and "Home Address" in Questions 1 and 2 should be removed to ensure respondent anonymity [1 mark].
    - Requesting such details is unnecessary for the investigation's aim and may deter participation, particularly among elderly residents who may be cautious of potential scams [1 additional mark].

- In Section B, the Likert scale could be improved by adding a "Not Applicable" or "Unaware" option [1 mark].
    - This ensures that respondents who have never heard of the initiative are not forced to pick "Neutral," which would be inaccurate. [1 additional mark]. This would also distort the data when calculating average awareness levels.

- Subjective options like "Often" and "Not often" in Question 9 should be replaced with specific frequencies [1 mark].
    - This is because respondents might interpret these options differently and they may thus choose different answers although they may have the exact same repair behaviour, thus causing the data to be inconsistent and unreliable. [1 additional mark].
    - For example, the students could use options like "Once a month" or "Every time a household item breaks," which makes the data more measurable and reliable for analysis [1 additional mark].

- The phrasing in Question 10 "Given that Repair Kopitiam is a fantastic initiative…" should be changed to a neutral statement, such as "How likely are you to participate in a Repair Kopitiam session in the future?" [1 mark].
    - This is because the original wording acts as a leading statement, which might influence or pressure the respondent to give a positive answer. By using neutral language, the students ensure the data is more accurate and reflects the person's true opinion [1 additional mark].

(d) With reference to Resource 3, describe the relationship between awareness of the Repair Kopitiam initiative and frequency of repair by users of Our Tampines Hub. [4]

Award 1 mark for each description of the relationship between awareness of the Repair Kopitiam initiative and frequency of repair by users of Our Tampines Hub, to a maximum of 5 marks.

Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- There is a general positive relationship between the level of awareness of the Repair Kopitiam initiative and the frequency of repair behaviours among users of Our Tampines Hub. [1 mark]

- As the awareness rating increases from 1 to 5, the recorded frequency of repair behaviours generally rises from 1 to 4. [1 mark]
    - For instance, two respondents with the lowest awareness level (Rating 1) reported the lowest frequency of repair (Rating 1). **Or** All respondents with the highest awareness (Rating 5) reported a high repair frequency of 3 or 4. [1 additional mark]

- The linear trend line indicates a moderately strong relationship between the two variables, as seen by the relatively high correlation coefficient (r-value) of 0.64. [1 mark]

- However, there is an anomaly - one respondent with low awareness (Rating 2) still reported a maximum repair frequency of 4. [1 mark]

(e) The students concluded that awareness of the Repair Kopitiam initiative has contributed to a higher frequency of climate-mitigating repair behaviours among users of Our Tampines Hub. Evaluate the validity of their conclusion. [10]

Possible Approaches:

Candidates could evaluate the investigation by reviewing the methods used to collect data and the data represented in Resource 3 (the scatter graph).

For data collection methods, candidates could review the sampling strategy (including choice of location to conduct the investigation) to assess the impact on the **representativeness and reliability** of the data:

- Candidates may argue that surveying general hub users (e.g., library-goers or diners) in the Festive Plaza instead of the individuals actually participating in the repair session at Festive Drive reduces the validity of the investigation.
    - This is because the survey captures what people know (awareness) but fails to capture if they really have a practice of repairing their broken items. By not interviewing people at the coaching station, the students likely missed out on people who already had the habit of repairing.
- Candidates could critique the lunchtime 2-hour window and the small sample size (22 respondents), noting that this high margin of error makes it difficult to draw a definitive conclusion about the thousands of Sunday OTH users.
    - By only surveying for two hours during lunch (12:00 pm – 2:00 pm), the students only captured a specific "snapshot" of people, which is not representative of all users of OTH.
    - Moreover, the timing could also have explained why there was such a small sample size. Users at OTH during lunchtime are likely there for a specific reason—to eat at the hawker center or run quick errands. They are often busy and less willing to stop for a survey, which explains why the students struggled to reach their target sample size.

For the data collected in Resource 2 (Questionnaire), candidates could consider the design flaws that affect **accuracy and reliability**:

- Candidates could critique Question 9's use of subjective terms like “Often” or “Not often.” They should explain how differing interpretations by respondents make the data inconsistent.
- Candidates should identify Question 10 as a leading statement ("Repair Kopitiam is a fantastic initiative..."). They should evaluate how this pressures respondents to give a positive answer, distorting the accuracy of the "likelihood" of future repair behaviour.
- Candidates could note that requesting Full Names and Home Addresses is intrusive and likely deterred participation, especially among cautious elderly residents, further impacting the sample quality.

For the findings in Resource 3 (Scatter Graph), candidates could evaluate the strength of the conclusion:

- The scatter graph demonstrates a general positive relationship, supported by a correlation coefficient (r-value) of 0.64. This suggests that as awareness of the initiative increases, the frequency of climate-mitigating repair behaviours also tends to rise. → supports the validity of the conclusion.
- However, candidates could argue that awareness of a specific initiative may not be the cause of the behavior; seniors may repair items out of habit or frugality regardless of the initiative.

- Small sample size of 22 responses may also not be representative of all OTH users and thus lowers validity of the conclusion.

Levels marked using Generic Level Descriptors for H1 fieldwork evaluative questions.

Section B

Cluster 1 Sustainable Future and Climate Change

2 Bangkok is the capital city of Thailand, a less economically developed country. The city registered 9.1 million residents as of 2024.

Resource 4 shows waste generated in Bangkok, Thailand from 2003 to 2020. Resource 5 shows plastic leakage hotspots in a district in Bangkok, Thailand. Plastic leakage refers to plastic waste that escapes waste management systems in an area and accumulates in the surrounding environment. Resource 6A shows photographs of the Benjakitti Forest Park, a recently completed urban regeneration project located in central Bangkok. Resource 6B is an article on the Benjakitti Forest Park.

(a) Describe the trends in waste generation in Bangkok, Thailand from 2003 to 2020 shown in Resource 4. [4]

Award 1 mark for each description of a trend in waste generation, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- Generally, municipal solid waste (MSW) generated has increased slightly by 150 tonnes/day from 2003 to 2020, from 9350 tonnes/day to 9500 tonnes/day. [1 mark]
    - While MSW dipped slightly by 860 tonnes/day from 2003 to 2006, from 9350 tonnes/day to 8490 tonnes/day, it saw an increase from 2007 onwards till 2020. [1 additional mark]

- Food waste increased steadily across the years from 2003 to 2020 by 2839 tonnes/day, from 2860 tonnes/day to 5669 tonnes/day. [1 mark]

- Plastic waste saw an increase of 1333 tonnes/day from 2003 to 2007, from 2167 tonnes/day to 3500 tonnes/day. [1 mark]
    - Plastic waste generation remained generally stable until 2019, hovering between 2167 tonnes/day to 2242 tonnes/day, before seeing a sharp spike between 2019 and 2020 to 3500 tonnes/day. [1 additional mark]

(b) With reference to Resources 4 and 5, suggest why waste management is important for Bangkok, Thailand. [4]

Award 1 mark for each reason why waste management is important, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each reason, where applicable.

Possible responses include:

- Effective waste management is important as Bangkok generates a large quantity of plastic waste that could have negative environmental impacts if allowed to be left to accumulate OR effective waste management is necessary to prevent the negative environmental impacts that stem from lack of management of plastic waste. [1 mark]
    - Resource 4 shows a large volume of MSW generated at 9500 tonnes/day in 2020, and together with that, Resource 5 shows plastic leakage hotspots near or directly next to the Bangbua Canal, such as at the Roy Klong community with a high plastic waste leakage of 680 tonnes/year. This indicates that large amounts of plastic waste and other waste will enter the water system, resulting in water pollution and contamination of the water source. [1 additional mark]

- The increasing volume of food waste generation needs to be managed to prevent negative health impacts that stem from lack of management of accumulated food waste. [1 mark]
    - As seen in Resource 4, food waste increased steadily by 2839 tonnes/day from 2003 to 2020. If left unmanaged, the high volume and rapid increase in volume of food waste generation could lead to issues such as food waste left in open areas becoming a breeding ground for pests leading to the spread of disease vectors. [1 additional mark]

(c) Suggest strategies that cities like Bangkok, Thailand can implement to manage waste sustainably. [4]

Award 1 mark for each strategy suggested, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each strategy, where applicable.

Possible responses include:

- Cities like Bangkok can implement local-scale recycling programmes to reduce the high volume of waste generated and left unmanaged. [1 mark]
    - As a city in a less economically developed country, Bangkok may lack the financial resources for large-scale waste infrastructure. Hence, formalising low-cost recycling programmes in partnership with its existing informal economy, such as waste pickers, provides a feasible way to recycle waste and reduce the volume of unmanaged waste over time. [1 additional mark]

- Cities like Bangkok can encourage composting of food waste to manage the large and growing volume of food waste. [1 mark]
    - Cities with rapidly growing populations like Bangkok are likely to generate high and increasing volumes of food waste. Hence, community-level composting programmes could be introduced in residential areas to convert food waste into compost for Bangkok's urban farms or green spaces, reducing the volume of food waste sent to landfills and the associated health risks. [1 additional mark]

(d) Explain why urban reimaging is needed for cities in less-developed countries. [4]

Award 1 mark for each reason why urban reimaging is needed, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each reason, where applicable.

Possible responses include:

- Urban reimaging is needed for cities in less-developed countries to increase their global competitiveness and attract foreign direct investment (FDI). [1 mark]
    - Cities in less-developed countries often suffer from negative perceptions of poor public infrastructure and inadequate public services, which may deter investors. Hence, urban reimaging projects such as the development of flagship projects demonstrate these cities' potential and can help rebrand these cities as attractive destinations for investment. [1 additional mark]

- Urban reimaging projects can also promote sustainable urban development in cities in less-developed countries. [1 mark]
    - Many cities in less-developed countries often face negative social, economic and environmental impacts due to high rates of urbanisation, such as inadequate sanitation facilities and the proliferation of low-cost slum housing. Hence, urban reimaging projects such as those that involve greening the city can enhance residents' quality of life through improving social well-being and environmental quality, thus contributing to sustainable urban development in the long-term. [1 additional mark]

(e) With reference to Resource 6, explain how the Benjakitti Forest Park contributes to the sustainable urban development of Bangkok, Thailand. [6]

Award 1 mark for each explanation of how the park contributes to sustainable urban development, to a maximum of 6 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- The Benjakitti Forest Park contributes to the environmental dimension of sustainable urban development by providing habitat for wildlife that provide vital ecosystem services. [1 mark]
    - Resource 6 states that "91 bird species recorded in this rewilded urban landscape", demonstrating how the park has successfully created a habitat and restored biodiversity in the heart of Bangkok, contributing to long-term environmental sustainability. [1 additional mark]

- The park also contributes to the environmental dimension of sustainable urban development through its function in carbon sequestration to mitigate climate change effects. [1 mark]
    - The "hundreds of mounded islands" and "tropical foliage" within the park serve as carbon sinks, helping to offset Bangkok's urban carbon emissions and mitigate the effects of climate change in the long term. [1 additional mark]

- The park contributes to the social dimension of sustainable urban development through flood regulation, reducing the negative impacts of flooding on Bangkok's residents. [1 mark]
    - Resource 6 notes that the park features "four lakes capable of holding a million cubic metres of stormwater", and that "the park and its vicinity remained unaffected" during a severe monsoon season when much of Bangkok flooded, demonstrating the park's efficacy in protecting residents from flood-related injuries, displacement, and loss of life. [1 additional mark]

- The park also contributes to the social dimension of sustainable urban development by improving residents' social and mental well-being through the provision of recreational and leisure spaces. [1 mark]
    - As "the largest central park in inner Bangkok", the park attracts visitors for activities "ranging from jogging and cycling to family gatherings and cultural celebrations" that promote social interaction, while the "skywalk and multiple boardwalks" enhance accessibility, providing green spaces that support mental well-being in a dense urban environment. [1 additional mark]

- The park contributes to the economic dimension of sustainable urban development by attracting tourist spending. [1 mark]
    - Resource 6 notes that the park attracts "a large number of local and foreign visitors daily", and that visitors can "engage paid local guides for a tour of the Park", generating income for the local economy and supporting livelihoods. [1 additional mark]

3 The Sahel is a semi-arid region in Northern Africa bordering the Sahara Desert, and includes less economically developed countries such as Senegal, Mali, Niger, and Nigeria. The population of the Sahel is rapidly increasing, projected to reach over 500 million in 2050.

Resource 7 shows the global distribution of areas of greatest concern from climate-related impacts of global warming. Resource 8 shows annual precipitation anomalies for the Sahel from 1915 to 2015. Resource 9 shows the projected impact of climate change on cereal productivity in 2080, as a percentage change from 2000. Resource 10 is an article on the impact of climate change on malaria transmission in African countries.

(a) With reference to Resource 7, describe the global distribution of areas of greatest concern from climate-related impacts of global warming. [4]

Award 1 mark for each description of the global distribution, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- Most continents of the world generally experience at least 1 climate-related impact, such as North America, South America, Africa, Australia, Europe and Asia. [1 mark]
    - The most frequently experienced climate-related impact is extreme heat stress risk, experienced across all regions with at least 1 climate-related impact. [1 additional mark]

- The range of climate-related impacts experienced is the greatest in equatorial regions and generally decreases polewards. [1 mark]
    - Areas with 3 or more climate-related impacts are concentrated around the equatorial areas such as the northern region of South America and Africa and some parts of Asia and Southeast Asia, while northern parts of North America and Europe and the southern region of Australia experience only 1 climate-related impact or none at all. [1 additional mark]

- River flooding is absent as a climate-related impact in certain regions, including Australia, western North America, and southern Africa. [1 mark]

(b) With reference to Resource 8, describe the differences in trends in annual precipitation anomaly for the Sahel between the period of 1915 to 1965 and the period of 1966 to 2015. [4]

Award 1 mark for each description of a difference in trends, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- From 1915 to 1965, the Sahel experienced an overall positive annual precipitation anomaly, whereas from 1966 to 2015, the Sahel experienced an overall negative annual precipitation anomaly. [1 mark]
    - For most years within the time frame of 1915 to 1965, the temperature anomaly ranged from 0 cm to 2.8 cm/month. However, for most years within the time frame of 1966 to 2015, the temperature anomaly ranged from 0 cm to -3.4 cm/month.

- The degree of precipitation anomaly was greater in the period from 1966 to 2015 compared to 1915 to 1965. [1 mark]
    - From 1966 to 2015, the precipitation anomaly exceeded -3 cm/month twice, in 1982 and 1984, whereas from 1915 to 1965, the anomaly never exceeded 3 cm/month in either direction. [1 additional mark]

(c) Using Resources 8, 9 and 10, explain the impacts of climate change on the Sahel. [6]

Award 1 mark for each impact of climate change on the Sahel explained, to a maximum of 6 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- Climate change is likely to result in increased drought risk in the Sahel. [1 mark]
    - As shown in Resource 8, the Sahel has experienced an overall negative precipitation anomaly from 1966 to 2015, with anomalies exceeding -3 cm/month for 1982 and 1984. [1 additional mark]

- Climate change is projected to have a severe negative impact on cereal productivity in the Sahel by 2080. [1 mark]
    - Resource 9 shows that a significant part of the Sahel is projected to experience a decrease in cereal productivity of 50% or larger by 2080 relative to 2000 levels. [1 additional mark] OR
    - This would negatively impact the livelihoods of the large proportion of the population that is dependent on subsistence agriculture, exacerbating problems of food insecurity. [1 additional mark]

- Climate change is projected to worsen malaria transmission in the Sahel. [1 mark]
    - Resource 10 notes that climate change-related impacts are already experienced in the Sahel, such as in Niger where malaria cases “rose by 30% between 2019 and 2021, climbing from 2.19 million to 2.99 million cases”. [1 additional mark] OR
    - This would result in illnesses and even deaths within the population, especially as "children under five making up 75% of African malaria deaths" as stated in Resource 10. [1 additional mark]

(d) Suggest how countries like those located in the Sahel could respond to climate change. [4]

Award 1 mark for each strategy suggested, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each strategy, where applicable.

Possible responses include:

- Countries in the Sahel could develop and adopt drought-resistant crop varieties to adapt to the increasing likelihood of reduced crop yields due to climate change. [1 mark]
    - As a semi-arid region, the Sahel already experiences limited and rainfall, which is exacerbated by climate change. Hence, the use of drought-resistant crop varieties would increase the resilience of agriculture to warmer and drier conditions, helping to safeguard food security for the large proportion of the population dependent on subsistence farming. [1 additional mark]

- Countries in the Sahel could improve access to water stores and irrigation technology to help farmers overcome dry spells and droughts. [1 mark]
    - Given that the Sahel is a semi-arid region already prone to water scarcity, providing farmers with greater access to a range of water stores, such as rainwater harvesting systems and networks of pipes and canals could help them maintain crop yields during dry spells that would otherwise cause crop failure, reducing the risk of food insecurity. [1 additional mark]

- Countries in the Sahel could invest in early warning systems for extreme weather events such as droughts and floods to reduce the vulnerability of their populations to climate-related impacts. [1 mark]
    - As many countries in the Sahel are less economically developed, they may lack the financial and technical resources to respond effectively to extreme weather events after they occur. Hence, investing in early warning systems would allow governments and communities to prepare in advance, reducing loss of life and economic damage from climate-related events that are projected to increase in frequency and severity. [1 additional mark]

(e) Explain the political and economic challenges in implementing strategies to respond to climate change. [4]

Award 1 mark for each challenge explained, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each challenge, where applicable.

Possible responses include:

Economic Challenges

- Developing countries with limited financial resources may struggle to fund and deploy mitigation technologies, posing a significant economic challenge to implementing strategies to respond to climate change. [1 mark]
    - For example, technologies such as carbon capture and storage require substantial investment that many LDCs are unable to afford, hindering the ability of LDCs to implement such strategies. [1 additional mark]
    - Developing countries are likely to face financial barriers in implementing adaptation strategies, such as improving irrigation infrastructure or developing early warning systems, at an adequate scale to benefit its populace, limiting their effectiveness in building adaptation capacity. [1 additional mark]

Political Challenges

- Regions with political instability or weak governance may struggle to coordinate and implement effective climate actions. [1 mark]
    - Without stable institutions and effective governance, climate change policies may lack the consistency and enforcement needed to be effective over the long term, as changes in political leadership or ongoing conflict can disrupt the implementation of climate strategies. [1 additional mark]

- Climate change policies often require difficult trade-offs between short-term economic development goals and long-term climate benefits, making it politically challenging to garner public and business support. [1 mark]
    - Policies and actions taken will need to be done in consultation with citizens and businesses, who may hold differing views on how development should proceed and may be resistant to climate change policies that do not yield immediate returns. This can slow down or dilute the implementation of effective climate strategies. [1 additional mark]

- International climate negotiations are often complex and protracted, as different countries have different needs and interests, making it difficult to reach consensus on effective global climate action. [1 mark]
    - For example, the Kyoto Protocol established differing responsibilities for developed and developing countries, reflecting the difficulty of balancing competing national interests. The greater the number of stakeholders involved, the longer it takes to reach consensus, as various parties hold different views, value systems, and priorities, complicating and prolonging decision-making that prevents the timely implementation of strategies to respond to climate change. [1 additional mark]

- A lack of political will among major carbon emitters to comply with international agreements poses a significant challenge to the effectiveness of global climate action. [1 mark]

- For example, the United States, one of the largest carbon emitters in the world, withdrew from the Kyoto Protocol before it even came into force. Without the cooperation of major emitting countries, mitigation efforts are unlikely to be effective in reducing global greenhouse gas emissions in the long run. [1 additional mark]

Section C

4 Evaluate the extent to which strategies to improve the lives of slum dwellers have led to progress towards sustainable urban development. [13]

Possible Approaches:

Candidates could approach the question by making a judgement on whether some strategies aimed at improving the lives of slum dwellers have led to progress towards SUD through a consideration of the relative strengths and limitations of the strategies. The strategies could be compared in relation to criteria such as feasibility, comprehensiveness in addressing varied needs, and whether they address the root issues that hinder SUD. These strategies could be aimed at improving existing standards of living, resettling, provision of large-scale public housing, or site and services schemes.

Candidates could also approach the question by making a judgement on whether the same strategies to improve the lives of slum dwellers have been more effective in progressing towards SUD in some places than others through a consideration of two or more case studies. Candidates could evaluate whether the strategies in each case achieved the intended aims of SUD. Candidates could also analyse the unintended negative impacts of the strategies on SUD. Candidates could also analyse the contextual factors such as the scale of the issues, the political will of the authorities and the resources available, which may have contributed to the variations in progress between these places.

Levels marked using Generic Level Descriptors for H1 essays

5 "Cities can never be liveable for women."

Evaluate this statement. [13]

Possible Approaches:

Candidates could approach the question by making a judgement on the validity of the statement, through a consideration of the relative strengths and limitations of strategies implemented to address the issues faced by women in cities to improve the liveability of the city for them. Candidates could employ evaluative criteria to evaluate strategies aimed at addressing different dimensions of liveability for women, such as those related to economic well-being, social well-being, and psychological well-being. Candidates could also consider strategies implemented at different spatial scales or by different stakeholders.

Candidates could also approach the question by making a judgement on whether some cities have been more successful in improving its liveability for women than others, through a consideration of two or more case studies. Candidates could evaluate whether the strategies in each case achieved their intended aims and analyse any unintended negative impacts. Candidates could also analyse contextual factors such as the scale of the issues, the political will of the authorities, cultural attitudes towards gender equality, and the resources available, which may have contributed to variations in success between places.

Levels marked using Generic Level Descriptors for H1 essays.

6 "Human activities are the main culprits of contemporary climate change".

Evaluate this statement. [13]

Possible Approaches:

Candidates could approach the question by making a judgement on the validity of the statement by considering if contemporary climate change is largely caused by human factors (such as the burning of fossil fuels or deforestation that lead to increased concentration of greenhouse gases) or natural factors (including variations in solar output, the thermohaline circulation, etc.). Candidates could consider evidence of past climate variability and episodes of warming and cooling of Earth during the Quaternary period, and how far they correspond with variations in natural factors or changes in human activities over a similar period.

Candidate could also approach the question by making a judgement on whether human activities play a more significant role in exacerbating or mitigating contemporary climate change. Candidates may consider how human activities (such as deforestation and industrial activities) can contribute to recent global warming, and weigh this against the extent to which human activities (such as reforestation or artificial carbon sequestration) may control or slow down the rate of contemporary climate change. Candidates could analyse this in terms of spatial scale, temporal scale, or the (in)adequate responsibility and commitment of relevant actors.

Levels marked using Generic Level Descriptors for H1 essays.